LLM Security Gatekeeper

Detect adversarial prompts before they reach production LLMs

- Research pipeline for prompt injection + jailbreak detection
- Current repo state: classic ML, LLM/hybrid routing, DeBERTa fine-tuning, and external baseline benchmarking
- Goal: maximize adversarial catch-rate without making benign UX unusable

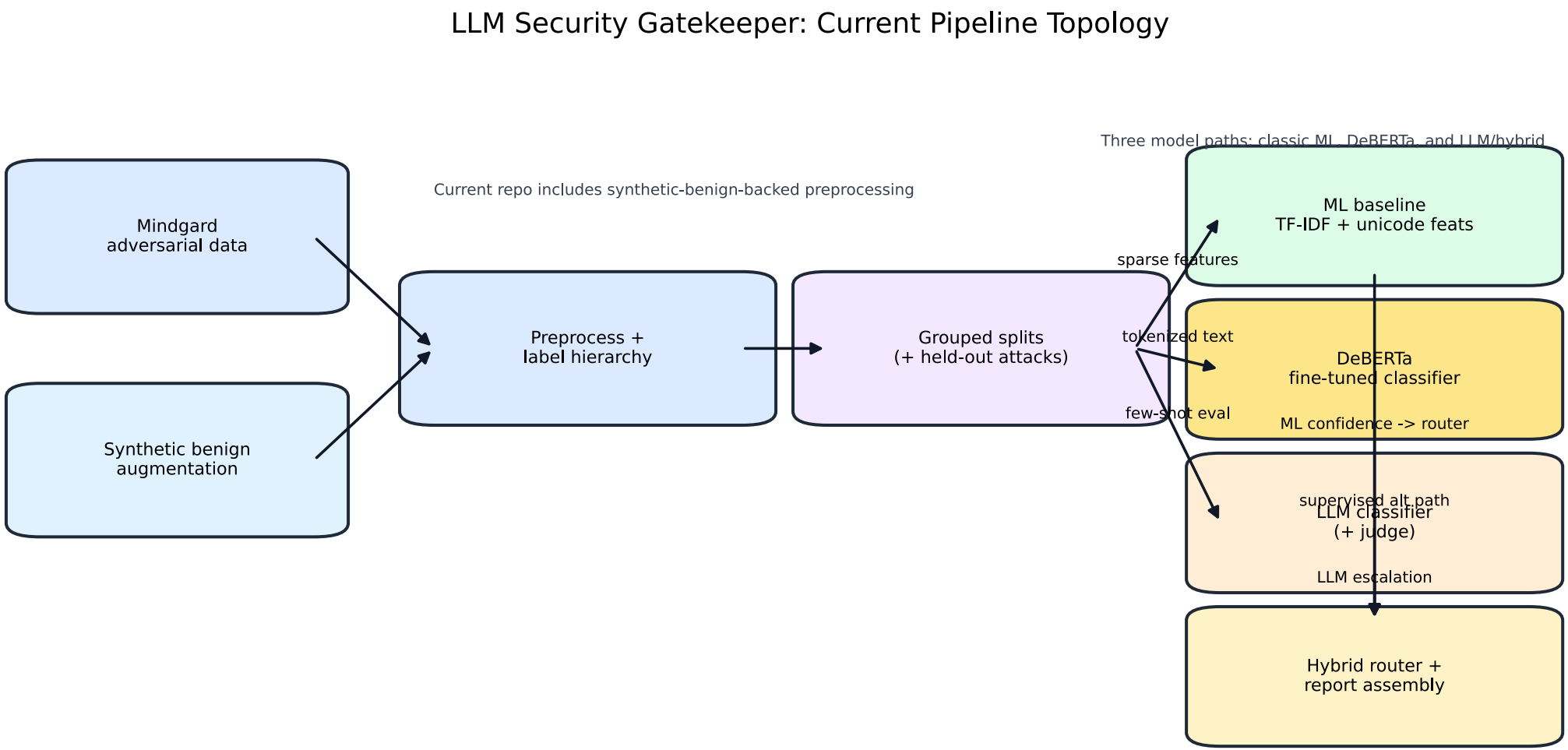
The Problem

- We need to classify prompts as `adversarial` vs `benign` before model execution
- Missed attacks are direct security failures
- False positives still matter:
  - they block legitimate user intent
  - they reduce trust in the guardrail
- The project also tracks attack category/type for analysis, not just block/allow

Why This Is Hard

- Attackers deliberately obfuscate prompts:
  - unicode tricks
  - paraphrase / typo / jailbreak variants
- Distribution shift is severe across public external datasets
- The data is highly imbalanced toward adversarial samples
- A production system needs a latency/cost-aware policy, not just one best offline score

Current Pipeline



DVC pipeline now includes synthetic-benign-backed preprocessing

Data Construction and Coverage

Dataset/Split	Rows	Adversarial	Benign	Adv %
train	7,989	7,352	637	92.0%
val	1,454	1,318	136	90.6%
test	1,819	1,682	137	92.5%
test_unseen	820	820	0	100.0%

- Benigns are no longer just de-duplicated originals; synthetic benign data is configured into preprocessing
- Held-out attack types still support unseen-attack testing
- External datasets vary sharply in size and class balance, including a very large `spml` stress set

Data Coverage and Class Balance

Modeling Paths

- ML baseline:
  - TF-IDF char n-grams + handcrafted unicode features
  - strong specialist for unicode-heavy attack patterns
- LLM + hybrid:
  - LLM classifier/judge path
  - hybrid router uses ML confidence to decide escalation
- DeBERTa:
  - newer supervised path in the repo for binary classification
  - trained and evaluated through its own DVC stage / CLI

DeBERTa Update

DeBERTa Training Update

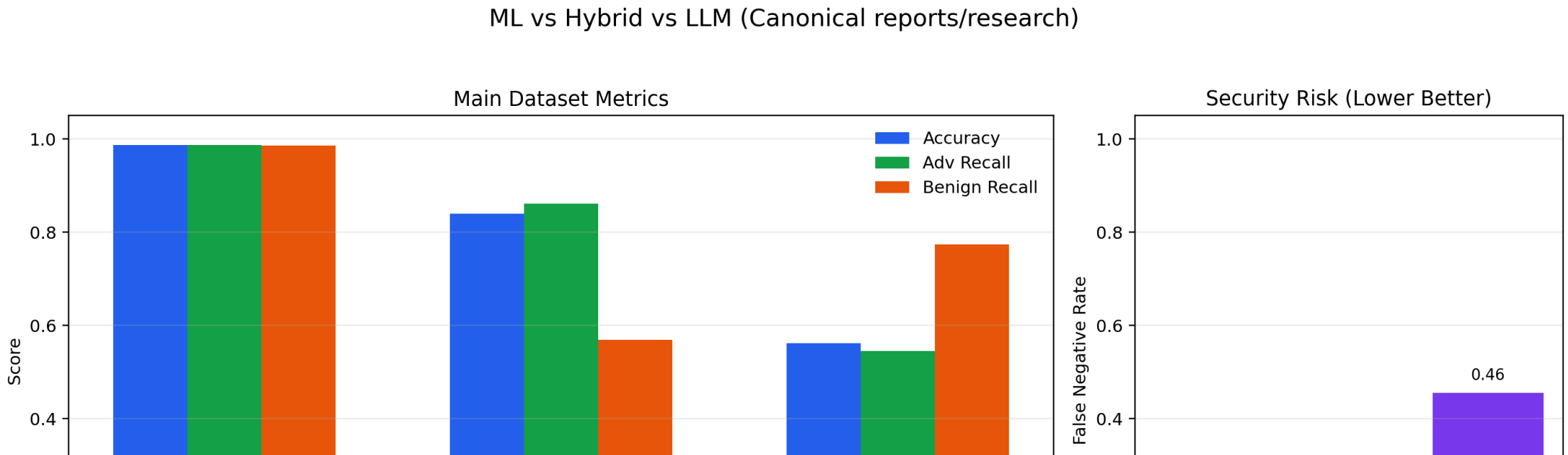
Model: microsoft/deberta-v3-base

Recent repo changes

- Class-weighted cross-entropy for severe class imbalance
- Longer schedule: 15 epochs, batch size 16
- Learning rate raised to 5.0e-06; patience 3
- Best-checkpoint save/restore plus WandB logging in research runs
- Current best checkpoint: epoch 2, F1 0.971

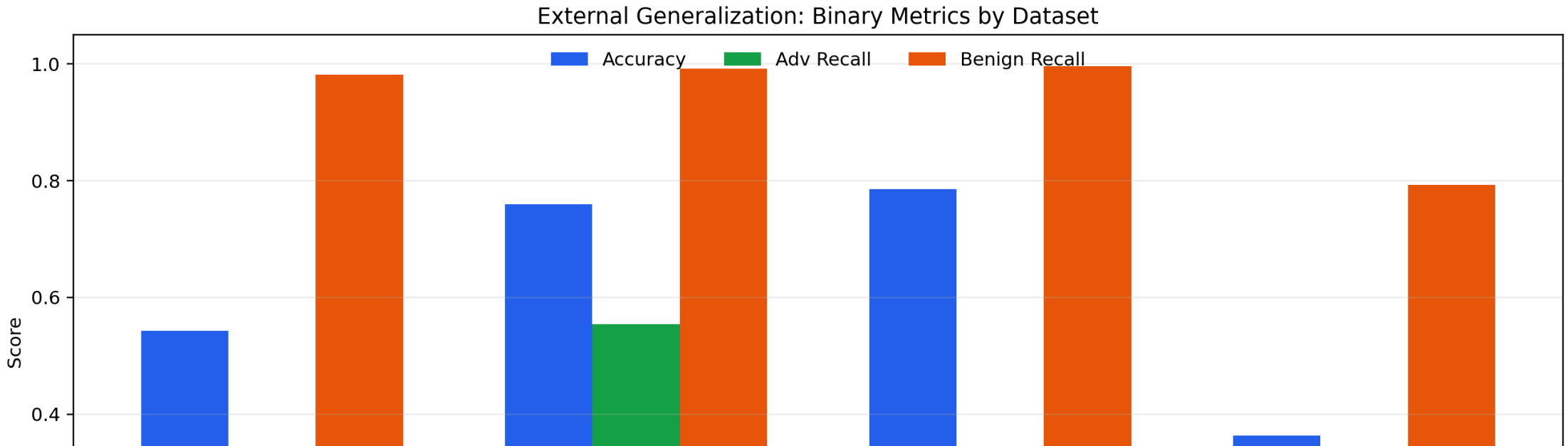
Main Results From Canonical Reports

Mode	n	Accuracy	Adv Recall	Benign Recall	FNR
ML (unicode scope)	1,094	0.9872	0.9875	0.9854	0.0125
Hybrid	1,819	0.8395	0.8615	0.5693	0.1385
LLM	1,819	0.5618	0.5446	0.7737	0.4554



External Generalization Is Still The Main Risk

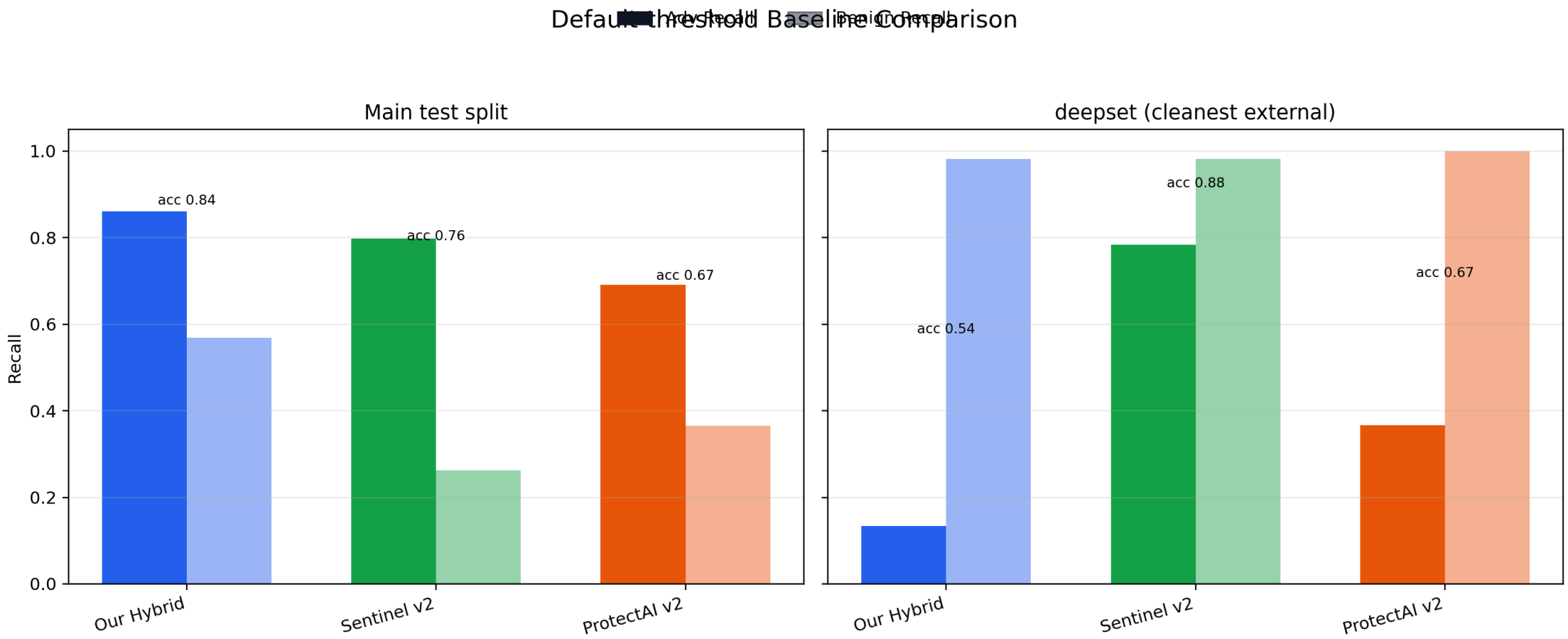
Dataset	n	Accuracy	Adv Recall	Benign Recall	FNR
deepset	116	0.5431	0.1333	0.9821	0.8667
jackhhao	262	0.7595	0.5540	0.9919	0.4460
safeguard	2,049	0.7857	0.3302	0.9964	0.6698



External Benchmarks Need Caveats

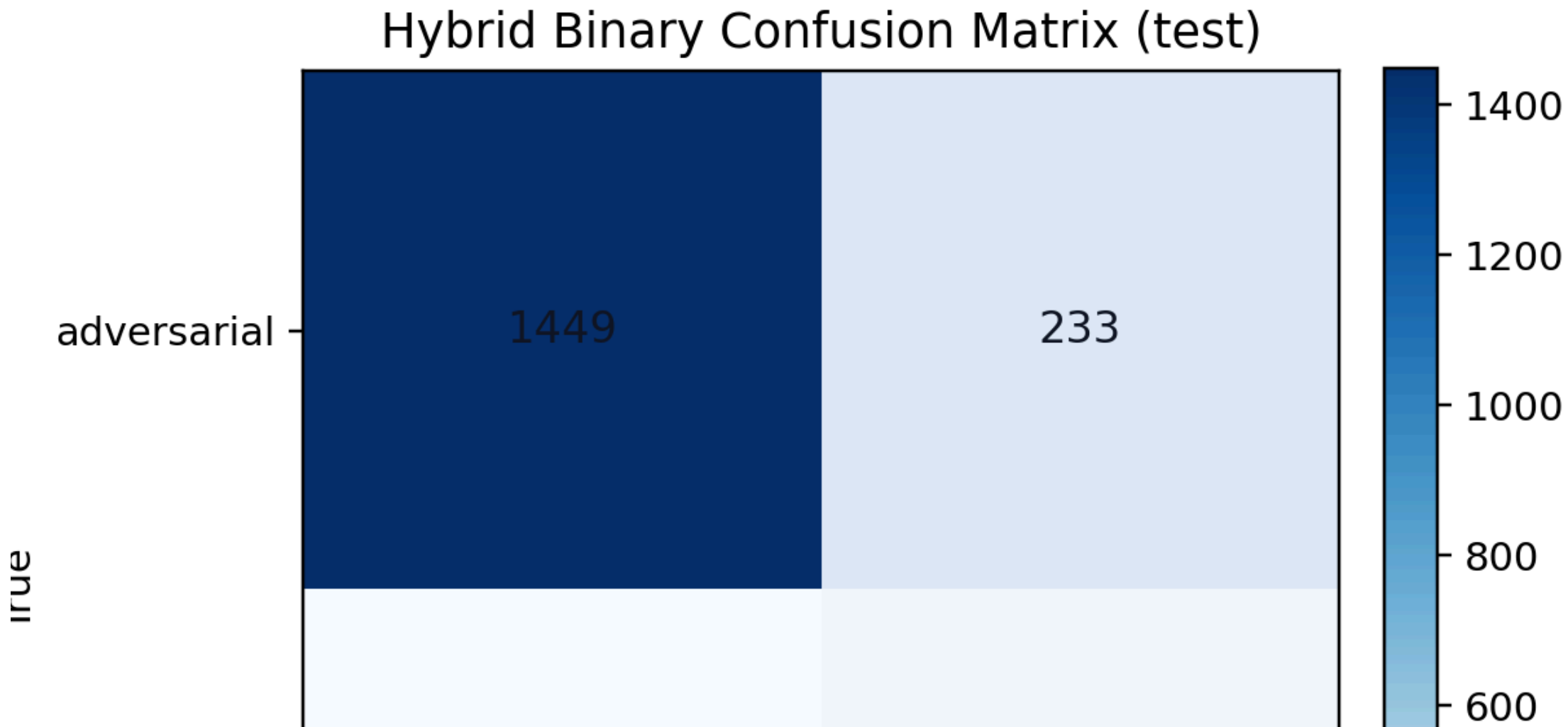
- `deepset` is the cleanest external benchmark in the repo today
- `jackhhao` has small exact overlap with our local corpus and is explicitly a ProtectAI v2 training source
- `safeguard` is not a clean unseen benchmark:
  - Mindgard is documented as Safe-Guard-derived
  - the repo overlap audit found large exact-text overlap
- So external numbers need provenance-aware interpretation, not just leaderboard reading

Baseline Comparison



- The repo now benchmarks against two public models:
  - Sentinel v2

Error Pattern



Production View

- The likely serving shape is still two-stage:
  - cheap local classifier first
  - slower escalation only when uncertainty or policy requires it
- The repo already supports many research ingredients needed for production:
  - routing stats
  - calibration analysis
  - reproducible report generation
- Still missing:
  - production API/policy engine
  - full latency/cost benchmarking
  - human review / audit workflow

What Changed Recently

- Synthetic benign generation is now part of the practical data story
- DeBERTa became a real first-party model path rather than an idea
- The project added external HuggingFace baseline comparisons
- The repo also now includes an overlap audit, which changes how to present “external” results responsibly

Next Steps

1. Improve benign realism/diversity and recalibrate routing
2. Push external adversarial recall up on the cleanest benchmark first (`deepset`)
3. Decide whether DeBERTa or a public baseline should become the primary low-latency classifier
4. Add production-facing evaluation: latency, cost, drift, and policy behavior

Appendix: Canonical Sources

- Current narrative should anchor on:
  - `reports/research/summary_report.md`
  - `reports/research/eval_report_{ml,llm,hybrid}.md`
  - `docs/2026-03-13_baseline_dataset_overlap.md`
  - `configs/default.yaml`
  - `dvc.yaml`
- DeBERTa state is backed by:
  - `src/models/deberta_classifier.py`
  - `src/cli/deberta_classifier.py`
  - `artifacts/deberta_classifier/best_checkpoint.json`